

Hao-Chun Liang

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EDUCATION

National Yang Ming Chiao Tung University <i>M.Eng. in IPSI (Chip Design and Automation Group) GPA: 4.3/4.3</i>	Sep. 2025 – Jun. 2027 (Exp.) <i>Hsinchu, Taiwan</i>
National Tsing Hua University <i>B.S. in Electrical Engineering GPA: 4.1/4.3 Distinction Graduation</i> Transferred from Dept. of Physics (Top 1 of 65; College of Science Elite Student Award).	Sep. 2020 – Jun. 2024 <i>Hsinchu, Taiwan</i>

Graduate Coursework (NYCU, all A+): IC Design Laboratory (ICLAB), DSP VLSI Design, Deep Learning, Stochastic Processes

Selected Undergraduate Coursework (NTHU): IC Design Lab (A+), VLSI Testing (A+), VLSI Design, Computer Architecture, Machine Learning, Digital Communication System (A+)

WORK EXPERIENCE

Digital IC Design Intern (Returning) <i>Andes Technology Corporation</i>	Jul. 2026 – Present <i>Hsinchu, Taiwan</i>
Graduate Research Assistant <i>National Yang Ming Chiao Tung University Parallel Computing System Laboratory</i>	Sep. 2025 – Jun. 2027 (Exp.) <i>Hsinchu, Taiwan</i>
Digital IC Design Intern <i>Andes Technology Corporation Acceleration R&D Division</i>	Nov. 2024 – Aug. 2025 <i>Hsinchu, Taiwan</i>

AWARDS & CERTIFICATES

Elite New Graduate Scholarship <i>Industry Academia Innovation School, NYCU</i>	Sep. 2025 <i>Hsinchu, Taiwan</i>
Excellent Poster Award <i>Taiwan and Japan Conference on Circuits and Systems (TJCAS)</i>	Aug. 2025 <i>Kaohsiung, Taiwan</i>
Cell-Based Digital Chip Design and Implementation (ADFP - TSMC 16nm) <i>Taiwan Semiconductor Research Institute (TSRI)</i>	Aug. 2024 <i>Hsinchu, Taiwan</i>
Distinction Graduation (Top 5% of 116 in Professional-Course GPA) <i>National Tsing Hua University Electrical Engineering</i>	May 2024 <i>Hsinchu, Taiwan</i>
College of Science Elite Student Award (Top 1 of 65) <i>National Tsing Hua University College of Science</i>	Apr. 2021 <i>Hsinchu, Taiwan</i>
Presidential Award (Received 3 times) <i>National Tsing Hua University</i>	2020 – 2022 <i>Hsinchu, Taiwan</i>

PUBLICATIONS

UniTrac: A Unified FPGA Accelerator for GNN-based Particle Track Reconstruction in HEPs <i>ACM Transactions on Reconfigurable Technology and Systems (In-Submission) Second + Corresponding Author</i>
Real-Time GPU Kalman-Filter Tracking via Kernel Refactoring and INT8 Surrogates <i>Fast Machine Learning for Science Conference 2025, ETH Zurich First Author</i>
An Integrated FPGA Implementation of Complete GNN-Based Trajectory Reconstruction <i>Taiwan and Japan Conference on Circuits and Systems (TJCAS) 2025 Second Author</i>
A High-Performance Implementation of GNN-Based Trajectory Reconstruction on FPGA <i>VLSI Design/CAD Symposium 2025 Second Author</i>
Fine-Grained Image Recognition from Scratch with Teacher-Guided Data Augmentation <i>arXiv:2507.12157 Fifth Author</i>

PROJECTS

Regfile Automatic Code Generation

Andes Technology Corporation

Nov. 2024 – Aug. 2025

Hsinchu, Taiwan

Ultra-Lightweight HDC-CNN for NeurIPS 2025 Weak Lensing Challenge

NeurIPS FAIR Universe Challenge

Oct. 2025 – Nov. 2025

Top-20 Ranking

NVIDIA NSYS/NCU Profile and Analysis of Token Reduction for ViT

National Yang Ming Chiao Tung University | Parallel Computing System Laboratory

Oct. 2025 – Nov. 2025

Hsinchu, Taiwan

RISC-V 5-Stage Pipelined CPU Core

National Tsing Hua University

Dec. 2023 – Jan. 2024

Hsinchu, Taiwan

SKILLS

Programming Languages: Python, C/C++, Verilog/SystemVerilog, HLS, CUDA, Perl, Shell

Domains: Efficient Machine Learning, Computer Architecture, Digital IC Design, RTL/C++ Codegen